

## **L3a:** Equity Exchanges, Growth Rates, and Stylized Facts

CHEME 5660 Cornell University

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# Objectives for Today

Today we connect equity-market infrastructure to price data, convert prices into growth rates and returns, and use recurring empirical patterns to test a model of equity behavior.

## Today's concepts

- **Describe equity claims:** distinguish the corporation, an exchange, a stock index, and an exchange-traded fund.
- **Compute growth rates:** calculate continuously compounded growth rates and convert them to log returns using  $r = (\Delta t)g$ .
- **Test return models:** use return distributions and autocorrelation to check whether a model matches observed market data.

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Companion notes: [L3a lecture notes and notebooks](#)

# Lectures and Examples

## **Lecture:**

CHEME-5660-L3a-Lecture-Equity-Exchanges-Return-StylizedFacts-Fall-2026.ipynb

**Example:** CHEME-5660-L3a-SVD-SP500-Example-Fall-2026.ipynb

Use SVD to identify common patterns in the growth rates of S&P 500 constituents.

**Example:** CHEME-5660-L3a-StylizedFacts-Example-Fall-2026.ipynb

Compare distribution fits, inspect tail-index stability, and contrast the memory of signed growth with the memory of growth magnitude.

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**Link:** <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

# Exchanges, Stocks, and ETFs

Before measuring an equity return, separate the marketplace from the securities traded there. These objects interact, but they are not interchangeable.

- An exchange is an organized venue that matches orders and imposes listing and trading rules.
- A stock is a residual ownership claim on a corporation.
- An index is a rule-based numerical portfolio benchmark. It is not itself a security.
- An exchange-traded fund, or ETF, is a security whose portfolio may seek to track an index and whose shares trade during the day.

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**Today:** a market price is the price available at a particular time and trading venue.

# Exchanges Connect Buyers and Sellers

An exchange brings orders together and supports **price discovery**, the process by which trading establishes a market price, and **liquidity**, the ability to trade without moving that price excessively.

- Several U.S. venues compete for the same order, including NYSE, Nasdaq, Cboe BZX, and IEX.
- A broker may route one order to a venue or divide it among venues, subject to available prices, liquidity, and the order instructions.
- After execution, separate clearing and settlement infrastructure completes the transfer of securities and funds.

The **National Best Bid and Offer**, or NBBO, reports the highest displayed bid and the lowest displayed ask across the national market system.

# Order Books and the NBBO

## Order books and the NBBO

National Best Bid and Offer

NASDAQ • XYZ

| SELL / ASK | SIZE | PRICE |
|------------|------|-------|
|------------|------|-------|

|  |     |       |
|--|-----|-------|
|  | 200 | 50.07 |
|--|-----|-------|

|  |     |       |
|--|-----|-------|
|  | 400 | 50.05 |
|--|-----|-------|

|      |     |       |
|------|-----|-------|
| BEST | 100 | 50.04 |
|------|-----|-------|

— spread = \$0.03 —

BUY / BID

|      |     |       |
|------|-----|-------|
| BEST | 300 | 50.01 |
|------|-----|-------|

|  |     |       |
|--|-----|-------|
|  | 500 | 50.00 |
|--|-----|-------|

|  |     |       |
|--|-----|-------|
|  | 200 | 49.99 |
|--|-----|-------|

TOP OF BOOK • XYZ

| VENUE | BID | ASK | SPREAD |
|-------|-----|-----|--------|
|-------|-----|-----|--------|

|      |       |       |      |
|------|-------|-------|------|
| NYSE | 50.00 | 50.05 | 0.05 |
|------|-------|-------|------|

|        |       |       |      |
|--------|-------|-------|------|
| Nasdaq | 50.01 | 50.04 | 0.03 |
|--------|-------|-------|------|

|          |       |       |      |
|----------|-------|-------|------|
| Cboe BZX | 50.02 | 50.06 | 0.04 |
|----------|-------|-------|------|

|     |       |       |      |
|-----|-------|-------|------|
| IEX | 50.01 | 50.06 | 0.05 |
|-----|-------|-------|------|

best bid and best ask sit  
on different venues

NBBO • XYZ

BEST BID

**\$50.02**

Cboe BZX

BEST ASK

**\$50.04**

Nasdaq

**spread \$0.02**

tighter than any single  
venue's own spread

# Stocks Are Ownership Claims

A stock, also called a share or **equity security**, represents fractional ownership in a corporation. It is a residual claim, not a promise of fixed repayment.

## What the shareholder may receive

- Voting rights and cash distributions such as dividends.
- A claim on remaining assets only after contractual creditors are paid.
- A capital gain if another investor later pays a higher share price.

Limited liability ordinarily caps the shareholder's loss at the invested amount. It does not protect the share price from falling to zero.

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**Source:** [U.S. Securities and Exchange Commission, Stocks](#)



# ETFs Turn Portfolios into Tradable Securities

An ETF pools assets into a portfolio and issues shares that trade on an exchange throughout the day. Each share represents an interest in that portfolio.

- A broad-market ETF can provide diversified exposure, but sector, leveraged, inverse, and single-stock ETFs may be highly concentrated.
- The **net asset value**, or NAV, is the fund's assets minus its liabilities, divided by shares outstanding.
- An ETF has both a market price and a NAV per share. They are usually close, but the ETF can trade at a premium or discount to NAV.

Like a stock, an ETF produces an observable sequence of market prices.

# From Securities to Price Data

Let  $S_t > 0$  denote the observed market price of a stock or ETF at time  $t$ . The statistic computed from that sequence depends on what the sequence means.

## Two data conventions

- A **price return** uses the quoted price series and omits cash distributions such as dividends.
- A **total return** includes distributions, commonly through an adjusted-price series whose vendor convention must be documented.

State the adjustment rule, sampling interval, and time units before comparing growth rates or risk statistics.

# Reward

We measure price reward with the **continuously compounded growth rate**, denoted by  $g$ . Let  $S_{j-1}^{(i)} > 0$  and  $S_j^{(i)} > 0$  be consecutive prices for firm  $i$ , separated by  $\Delta t_j > 0$  years. The price evolution is:

$$S_j^{(i)} = S_{j-1}^{(i)} \exp(g_{j,j-1}^{(i)} \Delta t_j).$$

Solving for the rate gives:

$$g_{j,j-1}^{(i)} = \underbrace{\frac{1}{\Delta t_j} \log \left( \frac{S_j^{(i)}}{S_{j-1}^{(i)}} \right)}_{\text{growth rate}}.$$

The growth rate has units of inverse years. Its sign records whether the price increased, decreased, or remained unchanged. The notebook uses closing prices, so this is a price-growth measure unless an adjusted series is selected.

# Growth Rates and Log Returns

Let  $g_t$  be the continuously compounded growth rate over an interval of length  $\Delta t_t > 0$  years. The corresponding **log return**  $r_t$  is dimensionless and is given by:

$$\underbrace{r_t = (\Delta t_t)g_t}_{\text{log return}} = \log\left(\frac{S_t}{S_{t-1}}\right).$$

Log returns add across adjacent intervals. For positive prices  $S_0, S_1, \dots, S_T$ , the cumulative log return is:

$$\sum_{t=1}^T r_t = \underbrace{\sum_{t=1}^T (\Delta t_t)g_t}_{\text{time-weighted growth}} = \log\left(\frac{S_T}{S_0}\right).$$

Use  $g$  to compare annualized rates. Convert to  $r$  before aggregating across time.

# Excess Growth Compares an Asset with a Benchmark

Let  $g_t$  be the equity growth rate and let  $g_{f,t}$  be a declared benchmark growth rate over the same interval. The **excess growth rate** and excess log return are given by:

$$\underbrace{\bar{g}_t = g_t - g_{f,t}}_{\text{excess growth}}, \quad \underbrace{\bar{r}_t = (\Delta t_t) \bar{g}_t}_{\text{excess log return}}.$$

- $\bar{g}_t > 0$ : the equity price grew faster than the benchmark.
- $\bar{g}_t < 0$ : the equity price grew more slowly than the benchmark.
- $\bar{g}_t = 0$ : both produced the same continuously compounded growth rate over the interval.

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**Careful:** the benchmark must use the same currency, horizon, and time units. A positive excess rate is not a guarantee of positive wealth.

# Algorithm

Let  $\mathcal{D}_i = \{S_1^{(i)}, \dots, S_T^{(i)}\}$  be a positive price series for firm  $i$ , let  $\Delta t > 0$  be the interval in years, and let  $g_f$  be the benchmark growth rate.

1. For  $t = 2, \dots, T$ , look up  $S_{t-1}^{(i)}$  and  $S_t^{(i)}$ .
2. Compute the excess growth observation:

$$\bar{g}_t^{(i)} \leftarrow \underbrace{\frac{1}{\Delta t} \log \left( \frac{S_t^{(i)}}{S_{t-1}^{(i)}} \right)}_{g_t^{(i)}} - g_f.$$

3. Store  $\bar{g}_t^{(i)}$ . The  $T$  prices produce  $T - 1$  observations.

Set  $g_f = 0$  when the required output is the growth rate rather than the excess growth rate.

# Common Patterns Across Many Assets

Let  $G \in \mathbb{R}^{m \times n}$  collect centered growth rates for  $m$  dates and  $n$  assets. Its singular value decomposition is:

$$G = U\Sigma V^T = \sum_{j=1}^{\text{rank}(G)} \underbrace{\sigma_j \mathbf{u}_j \mathbf{v}_j^T}_{\text{rank-one pattern}}, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq 0.$$

Large singular values identify common patterns that account for much of the variation across assets.

SVD finds patterns in the data; it does not explain their economic causes. Results depend on centering, scaling, missing-data treatment, and the selected assets.

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**Notebook:** CHEME-5660-L3a-SVD-SP500-Example-Fall-2026.ipynb

# Risk

Risk is uncertainty about loss or gain. A common dispersion measure is the sample standard deviation of the growth rate, denoted by  $\sigma_g$ .

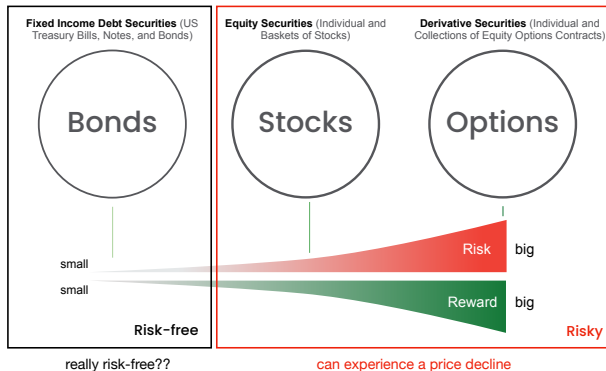
Suppose  $T$  prices produce the  $T - 1$  observations  $g_2, g_3, \dots, g_T$ . Let  $g'$  be their sample mean. The sample volatility is:

$$g' = \frac{1}{T-1} \sum_{t=2}^T g_t, \quad \sigma_g = \underbrace{\sqrt{\frac{1}{T-2} \sum_{t=2}^T (g_t - g')^2}}_{\text{growth-rate volatility}}.$$

The statistic has the same inverse-year units as  $g$ . It measures dispersion around the sample mean, not tail probability, drawdown, liquidity, or the magnitude of a specific loss.



# Return and Risk Measure Different Things



Assets with similar sample means and standard deviations can still have very different tails, drawdowns, dependence, and liquidity.

# Other Risk Measures

Choose a risk measure that matches the type of loss relevant to the decision.

- **Value at Risk**: a loss threshold exceeded with a stated small probability over a stated horizon.
- **Expected Shortfall**: the average loss beyond that threshold.
- **Maximum drawdown**: the largest peak-to-trough decline in value.
- **Downside deviation**: dispersion measured only below a declared target.
- **Beta**: sensitivity to a selected market benchmark.
- **Tracking error**: volatility of the return difference from a benchmark.

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**Ask:** what type of loss matters, over what horizon, and will this statistic detect it?

# Stylized Facts of Equity Growth Rates

A **stylized fact** is a recurring empirical pattern observed across many assets or markets. It is not an exact law for every ticker and sample. It is a pattern that a useful model should be able to reproduce.

- **Heavy tails**: large movements occur more often than a fitted Gaussian model predicts.
- **Weak linear autocorrelation**: signed growth rates often have little serial linear dependence at ordinary liquid-market horizons.
- **Volatility clustering**: large movement magnitudes occur near other large movements, and calm observations near other calm observations.

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**Sources:** [Cont \(2001\)](#); [Mandelbrot \(1963\)](#)

# Heavy Tailed Return Distributions

A power-law tail decays more slowly than an exponential tail. Let  $X$  be a random variable, let  $\alpha > 0$  be its **tail index**, and let  $L(x)$  be slowly varying. A two-sided power-law tail satisfies:

$$\mathbb{P}(|X| > x) \sim \underbrace{L(x)x^{-\alpha}}_{\text{tail probability}} \quad \text{as } x \rightarrow \infty.$$

A smaller  $\alpha$  means slower decay and a heavier tail. The classical strict thresholds identify moments that fail to exist for a pure power-law tail:

- $\alpha < 1$ : the mean fails to exist.
- $\alpha < 2$ : the variance fails to exist.
- $\alpha < 4$ : the fourth moment fails to exist.

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**Careful:** at the boundary values 1, 2, and 4, moment existence depends on finer tail detail. Heavy-tailed does not always mean power-law.

# Hill's Estimator of the Tail Index

Hill's estimator applies to positive observations. For signed growth rates, analyze one tail or the magnitude series  $|g|$ . Let  $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)} > 0$  be descending order statistics, and let  $k < n$  select the upper-order observations. The estimate is:

$$\hat{\alpha}_k = \underbrace{\left[ \frac{1}{k} \sum_{i=1}^k \log \left( \frac{X_{(i)}}{X_{(k+1)}} \right) \right]^{-1}}_{\text{estimated tail index}}.$$

The cutoff  $k$  balances bias and variance. One arbitrary cutoff is not a diagnostic: compute  $\hat{\alpha}_k$  over a range of  $k$  values and look for a stable region.

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**Companion example:** the stylized-facts notebook uses a Hill stability plot and treats  $k = \lfloor \sqrt{n} \rfloor$  as a reproducible reference, not a uniquely correct cutoff.

# Autocorrelation

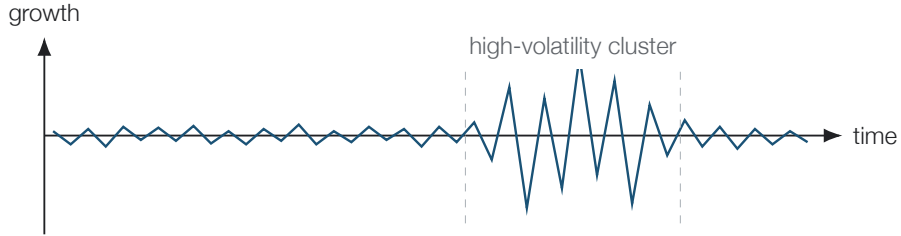
Let  $g_2, \dots, g_T$  be growth-rate observations with sample mean  $g'$ , and let  $k$  be a lag measured in sampling intervals. Define the empirical autocovariance and autocorrelation by:

$$\widehat{R}(k) = \frac{1}{T-1} \sum_{t=2}^{T-k} (g_t - g')(g_{t+k} - g'), \quad \underbrace{\rho(k) = \frac{\widehat{R}(k)}{\widehat{R}(0)}}_{\text{autocorrelation}}.$$

The normalization gives  $\rho(0) = 1$ . For many equity samples,  $\rho(k)$  is near zero at nonzero lags: past signed growth provides little **linear** help in predicting future signed growth.

Near-zero autocorrelation does **not** establish independence. Dependence can remain in growth magnitude, trading microstructure, or other nonlinear functions of the series.

# Volatility Clustering



**Volatility clustering** means large movements tend to occur near other large movements, and small movements near other small movements. The cluster contains both signs, so the dependence concerns magnitude rather than a persistent direction.

# Diagnosing Volatility Clustering

Let  $g_2^{(i)}, \dots, g_T^{(i)}$  be the growth-rate series for firm  $i$ , and let  $\overline{(g^2)}_i$  be the mean squared growth rate. The empirical autocorrelation of squared growth rates is:

$$\hat{\rho}_i(k) = \frac{\sum_{t=2}^{T-k} [(g_t^{(i)})^2 - \overline{(g^2)}_i] [(g_{t+k}^{(i)})^2 - \overline{(g^2)}_i]}{\sum_{t=2}^T [(g_t^{(i)})^2 - \overline{(g^2)}_i]^2}.$$

Volatility clustering appears when  $\hat{\rho}_i(k) > 0$  across many lags and fades slowly toward zero. A model can therefore match weak signed-growth ACF and still fail because it assumes fixed, independent volatility.

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**Why it matters:** conditional volatility changes risk forecasts, option values, hedge ratios, and capital allocation.



# Example

## Diagnosing Stylized Facts in Daily Equity Growth Rates

The companion notebook turns each stylized fact into a model check:

- Compare empirical growth rates with fitted Normal, Laplace, and Student- $t$  densities.
- Inspect a Hill stability plot instead of reporting one cutoff.
- Compare the ACF of signed growth with the ACF of squared growth.
- Translate the diagnostics into requirements for the next model.

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**Notebook:** `CHEME-5660-L3a-StylizedFacts-Example-Fall-2026.ipynb`

# Summary

This lecture introduced equity securities, continuously compounded growth rates, and recurring empirical patterns in financial return data.

## Key takeaways

- **Market infrastructure and securities are different.** Exchanges discover prices; stocks and ETFs are claims traded there.
- **Return statistics require conventions.** The growth rate annualizes and the log return aggregates. State price versus total return.
- **Stylized facts are model requirements.** Heavy tails and clustered volatility expose failures of a fixed-variance independent model.

These diagnostics become tests for the models that follow.

That's a wrap. What are some of the interesting things we discussed today?